Boyang Xu

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Summary

PhD Student in Data Science developing **generative models (flow matching, diffusion)** for **distributional reinforcement learning (RL) and multimodal machine learning**. I build generative models for **risk-aware** forecasting, predicting the full distribution of potential outcomes to improve **offline RL strategic decisions** (First author, **ICML 2026**). I have also deployed multimodal fusion models for a DOE-funded energy system **(IEEE CASE 2025 Best Paper, IISE DAIS 2023 Student Finalist)** and developed resource-efficient algorithms for NSF-funded disaster response.

Education

Arizona State University, *PhD in Data Science* **Expected May 2027**
Northeastern University, *MS in Electrical and Computer Engineering* **Jun 2022**
University of Jinan, *BS in Optoelectronic Information Science* **Jun 2020**

Technical Skills

Generative Models: Flow Matching, Diffusion Models
Multimodal Machine Learning: Time-Series Forecasting (iTransformer), Temporal Event Modeling, Cross-Modal Attention, Probabilistic Forecasting
Decision-Making: Offline & Distributional RL, POMDPs, Interpretable HMMs/IOHMMs, Bayesian Optimization
Tools: JAX, PyTorch, Python, Linux

Research Projects

Generative AI for Risk-Sensitive Reinforcement Learning Fall 2025-Fall 2026
Built a continuous-time generative AI framework for Reinforcement Learning (RL) that predicts the full probability distribution of decision outcomes, enabling safer, risk-aware actions in unpredictable autonomous systems.

- **Achieved best or top 5% performance among 7 leading methods across 38 offline RL tasks** by engineering a generative flow-matching model to predict full probability distributions of alternative actions.
- **Reduced distributional error by 66%** over prior state-of-the-art by innovating a shared-noise path coupling mechanism and parameterized control variate.
- **Engineered an open-source JAX benchmarking pipeline** evaluating 6 baselines at up to 1M steps per run on A100 GPUs, resulting in a first authored paper at ICML 2026 and an INFORMS 2026 talk.

Multimodal Event and Sensor Generative Modeling Fall 2023-Fall 2025
Built a multimodal forecasting model for a DOE-funded solar plant to accurately predict equipment failures and simulate future system health.

- **Achieved a 0.98 weighted F1 score for next-event fault prediction, improving upon a 0.77 baseline,** by fusing continuous sensor data and discrete event encoders (iTransformer and Transformer Hawkes Process) through a dynamic co-attention mechanism.
- **Reduced power prediction mean squared error (MSE) by nearly 44%** by simulating multi-day plant futures across 200 modules and 61 variables using an MLP decoder and an iterative rolling-window generation framework.

AI-Driven Autonomous Routing for UAV Assessment Fall 2023-Fall 2025
Created a resource-efficient active sensing algorithm for UAVs to rapidly and accurately assess disaster damage under strict flight-time constraints.

- **Achieved top predictive accuracy (85%) and weighted F1 score (0.81) on a 1,000-building tsunami simulation** by developing an Ordinal Deep Kernel Gaussian Process (ODGP) surrogate model that maps continuous risk scores to discrete, naturally ordered damage levels.
- **Reduced cumulative UAV travel costs by over 77% (14.69 vs. 64.14) compared to standard acquisition baselines** through a dynamic cost-aware level-set acquisition function that balances uncertainty exploration, critical boundary exploitation, and spatial travel penalization.
- Ensured **scalable, real-time trajectory optimization** by training the model via sparse variational inference (SVGP), allowing the agent to strategically target severely damaged regions while avoiding unnecessary flight time.

Selected Publications

- **B. Xu***, Q. Zou*, S. Yang, H. Yan. "Path-Coupled Bellman Flows for Distributional Reinforcement Learning." **ICML 2026**
- J. Huang, **B. Xu**, Y. Liu, H. Yan. "Multi-modal Generative Modeling of Event Sequence and Time Series for Solar PV Systems." Extended version to appear in *IEEE TASE*. **IEEE CASE 2025, Best Paper**
- **B. Xu**, M. Reisi Gahrooei, M. Ilbeigi, H. Yan. "Adaptive Sampling for Automated Post-Disaster Rapid Damage Assessment via Level-Set Cost-Aware Bayesian Optimization." **SIAM SDM 2026**
- **B. Xu**, Y. Kang, X. Zhao, H. Yan, F. Ju. "Partially Observable Markov Decision Process Framework for Operating Condition Optimization Using Real-Time Degradation Signals." **J. Quality Technology 2025**